

SACHIN SHEKHAWAT

B.Tech in Computer Science and Engineering | Indian Institute of Technology (ISM), Dhanbad



@ shekhawatsachin099@gmail.com +91 99822 30566 in [linkedin.com/in/sachin-shekhawat](https://www.linkedin.com/in/sachin-shekhawat) Pune, India

Software Engineer specializing in Java backend systems, distributed microservices, and payments infrastructure. Experienced in building scalable REST APIs, TDD-driven development, and cloud-native deployments across GCP, Docker, and Kubernetes.

EDUCATION

Bachelor of Technology, Computer Science 2021 – 2025
Indian Institute of Technology (ISM), Dhanbad
CGPA: 7.57 / 10

Coursework: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, **Distributed Systems, Database Management Systems**, Software Engineering, Probability & Stochastic Processes

EXPERIENCE

Barclays (India) Aug 2025 – Present
Technology Developer Pune, India

- Owned, as sole engineer, a Java Spring Boot + Angular Payments Migration Dashboard for SWIFT Messaging Interface (MI), enabling a live, zero-downtime enterprise migration of all applications from legacy SMS messaging to the high-throughput IMI-MT framework.
- Modeled the end-to-end SWIFT message routing domain (BIC, BICInstance, Logical Terminal, App, AppInstance, AppInstanceEnv, E2EFlow, Flow, FlowInstance) to represent MI, HOPS sanctions-screening, and Back Office applications in a single schema.
- Built real-time visualizations of incoming flows (SWIFT → BIC → Logical Terminal → MI → HOPS → Back Office) and outgoing ACK/NACK flows — comparable to UPI transaction acknowledgement patterns — including per-BIC Back Office distribution and combined incoming+outgoing E2E flow tracing.
- Implemented Maker-Checker approval workflows, soft-delete governance, and optimized indexing to support safe, auditable changes to production routing data during the migration.
- Secured REST APIs with JWT authentication, following a TDD approach with H2 for testing before migrating to Oracle DB with production-grade indexing.
- Classified downstream applications by type (MI, HOPS sanctions-screening, and Back Office) using AppInstanceEnv metadata alongside BIC country codes and Logical Terminals, letting the dashboard auto-derive per-BIC application counts and surface migration gaps in real time.
- Partnered with cross-functional application teams to validate the SMS-to-IMI-MT cutover in phases, using flow-level ACK/NACK reconciliation to catch discrepancies before they reached production traffic.

Barclays (United Kingdom) Jun 2024 – Aug 2024
Technology Developer Intern Northampton, United Kingdom

- Developed Java microservices for fraud strategy and analytics, boosting unit test coverage by 10% and testing efficiency by 15% to strengthen risk management workflows.
- Streamlined CI/CD automation and infrastructure deployments using Docker, Kubernetes, Jenkins, and OpenShift.
- Participated in code reviews and pair programming with senior engineers, applying feedback on Java best practices and design patterns to improve code quality and maintainability.

SKILLS

- Languages:** Java (17/21), C++, C, Python, Kotlin
- Backend & Cloud:** Spring Boot, Microservices, REST APIs, Docker, Kubernetes, GCP
- Databases & Testing:** PostgreSQL, OracleDB, MySQL, Redis, jOOQ, JUnit, TDD
- Tools:** Git, Jenkins, Linux, OpenShift, Jira

PROJECTS

Concurrent Restaurant Simulation [GitHub](#)

Tech: C, POSIX Processes, Shared Memory, Semaphores, Mutexes

- Modeled customers, waiters, and chefs as independent OS processes coordinating through shared memory segments and named POSIX semaphores to safely access shared kitchen and table resources.
- Designed mutex-protected critical sections with a strict resource-ordering strategy to eliminate race conditions and prevent deadlock under high-concurrency simulated loads.
- Built configurable simulation parameters (arrival rate, table count, kitchen capacity) to stress-test synchronization correctness across varying concurrency levels.

Convocation Portal [GitHub](#)

Tech: React.js (Next.js), Laravel, MySQL, REST APIs, MVC

- Led development of IIT (ISM)'s Convocation Portal, streamlining operations for 5,000+ students and cutting processing time by 40%.
- Designed scalable REST APIs and a normalized MySQL schema following MVC architecture, improving backend data retrieval performance by 30%.
- Built role-based workflows for students, faculty, and admins (registration, gown allocation, certificate issuance) with server-rendered Next.js pages for fast load times.

MedicoBert [GitHub](#)

Tech: Python, Flask, Google Cloud Vision API, BioBERT, LLaMA3, OCR

- Built a pipeline to extract structured medical information from handwritten prescriptions using OCR, LLaMA3, and fine-tuned BioBERT, centralizing patient-indexed records for efficient retrieval.